

POOP POWER!

Matthew Mazzotta is using Dog Poop to power the lights in a Massachusetts city park. Since dog owners anyway have to pick up after their pets, all they have to do is dump it in the the digester – an air-tight cylindrical container, where the dog faeces are broken down by anaerobic bacteria, and then used to give back to the grid.

<http://t.co/9fjkkSM>

ANDROID GAINING GROUND

According to comScore, Android gained 5 per cent market share in the US for the period of April to July 2010, while all other smartphone platforms declined. Even after the recent launch of iPhone 4 the iOS market share declined 1.3 per cent. Where have all the fanboys gone? One crappy antenna and you desert your favourite fruit?

<http://usat.ly/cqdw89>

BOEING FOLLOWS VIRGIN GALACTIC INTO SPACE

"Please ensure your spine is in the stiffened position and your fears of hurtling towards the stars are stowed for take-off. Any passengers not planning to visit the Space Station today should disembark now." Full coverage - NewZeland Herald. Still not affordable for any of us though.

<http://bit.ly/cnSPec>

ment register file reduces the need for large, complex out-of-order execution hardware. The physical register stores operands, so that uops only carry the pointers to the operands they need—a much lighter load. The savings here enable the use of Advanced Vector Extensions (AVX) instructions, which can use operands up to 256 bits in size and borrow 128 bits of the now–256-bit integer SIMD path, enabling twice the floating point throughput. (A look at the specs of the new physical register file is below.)

Because this increase in floating point performance introduces new demands, the memory cluster has been changed as well. There are now two load/store ports, and Sandy Bridge uses symmetric address ports so that each port can service a load or store address, resulting in double the load bandwidth. According to a presentation held Monday at IDF, this is the Sandy Bridge change that most increased performance: It results in a total of 48 bytes per cycle internal memory bandwidth.

Branch prediction

According to Intel, branch prediction is one of the most effective ways to increase processor performance, which is one of the reasons it's tweaked with almost every new CPU release. Sandy Bridge introduces a few changes that improve its accuracy even above the 5 percent usual increase.

First, it offers more targets. In a standard branch predictor, branches are marked as "taken" or "not taken" with either "strong" or

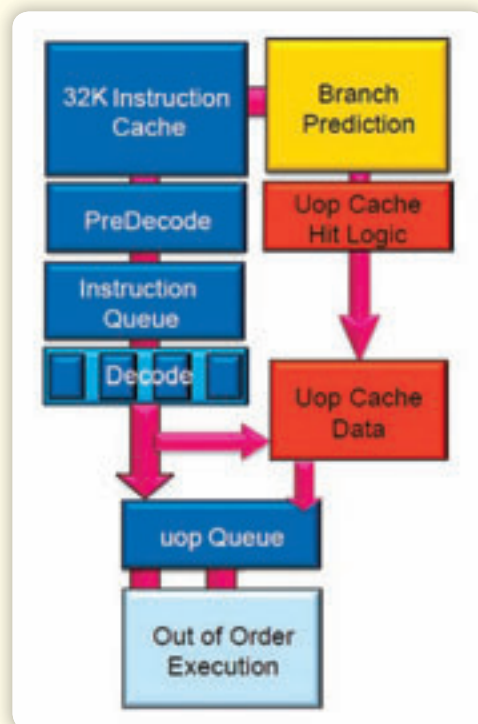
"weak" confidence. Intel representatives say they discovered that most branches predicted using that method, however, were marked with "strong" confidence, which diminished the accuracy of those predictions. With Sandy Bridge, Intel is moving to using one confidence bit for multiple branches, thus increasing the potential accuracy of any given one.

Second, branch targets were also changed, so that it's no longer a "one size fits all" situation. Instead there is support for multiple target sizes so that less space is wasted, the CPU can track more targets, and prediction speed can increase.

Third is the increase in the use of history bits, so that branches that need to look at long patterns of instructions and those that need to only look at shorter branches won't be grouped together.

One ring to unite them all

Zooming out, it's much easier to see one of Sandy Bridge's biggest innovations: uniting almost all of the major subsystems within the processor die itself. That's where you'll find the memory controller, the PCIe controller, and the graphics controller (which also use 32nm transistors—but more on them later). Because everything is snuggled so closely together, the processor needs a way to make sure everything gets along, using the available resources without utilizing too much power.



In Nehalem and Westmere, all cores had their own private path to the L3 cache. Sandy Bridge uses a ring-based interconnect, similar to the one used in Nehalem-EX and Westmere-EX, to link the cores, graphics, last level cache (see below), and System Agent. It contains four rings: the 32-byte data ring, the request ring, the acknowledge ring, and the snoop ring. They're all fully pipelined, so as the core count and cache size increase, so will the bandwidth to the cache. (This means it can also scale to servers using large numbers of processors.) As it's passing around the ring, data will always take the shortest path to minimize latency. Distributed arbitration and a sophisticated ring protocol handle coherency, ordering, and core interface issues.

Element	Nehalem	Sandy Bridge
Load Buffers	48	64
Store Buffers	32	36
RS - Scheduler Entries	36	54
PRF Integer	N/A	160
PRF float-point	160	144
ROB Entries	128	168